

ALEJANDRO TORRES CRUZ

Mechanical Engineering Student | CAD Modeling | CFD/FEA | Aerospace & Biomechanical Systems
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EDUCATION

University of Puerto Rico - Mayaguez | B.S. Mechanical Engineering

Expected May 2030

- GPA: 3.2/4.0 | Relevant Coursework: Engineering Graphics/CAD, Programming & Algorithms, Calculus, Physics

TECHNICAL SKILLS

CAD / Modeling Tools	SolidWorks, AutoCAD, PTC Creo, Onshape, Fusion 360, 3D part modeling, assembly modeling, mates, engineering drawings, tolerance-focused design, design for manufacturing
Simulation / Analysis Tools	ANSYS Fluent, ANSYS Mechanical, CFD setup, structural FEA, meshing, post-processing, numerical simulation
Programming	Python, NumPy, Matplotlib, numerical simulation, data analysis, iterative calculations
Technical Documentation	Engineering drawings, design reports, component documentation, manufacturing notes, research summaries, prototype improvement tracking

EXPERIENCE

Design Engineer, Colegio Prosthetics Research (CPR)

Dec 2025 - Jun 2026 | UPRM Biomedical Engineering Student Research Team | Mayaguez, PR | SolidWorks, CAD assemblies, engineering drawings

- Co-led a 7-member design/manufacturing team developing prosthetic hand components, including fingers, fingertips, palm geometry, wrist interfaces, and mechanical support systems.
- Designed and refined prosthetic finger, palm, and wrist-interface components in SolidWorks to improve ergonomics, component fit, realistic proportions, and manufacturability.
- Built and managed CAD assemblies by defining mates, component constraints, and key measurements to verify interaction between moving parts.
- Created engineering drawings for selected prosthetic components, defining critical dimensions, tolerances, and manufacturing details for fabrication and assembly.
- Led CAD skill-development workshops and created an end-of-semester documentation guide to standardize design decisions, tolerance notes, errors found, and future improvements.

ENGINEERING PROJECTS

Variable-Ramp Supersonic / Hypersonic Inlet CFD & FEA Study

2026

Independent Aerospace Design Project | SolidWorks, AutoCAD, ANSYS Fluent, ANSYS Mechanical

- Modeled a variable-ramp inlet assembly with inlet body, movable ramp, throat/isolator, diffuser duct, shaft, side brackets, and fasteners for a high-speed propulsion study.
- Built a preliminary Mach 6 CFD workflow to evaluate oblique shock behavior, velocity/Mach contours, static pressure trends, pressure losses, and possible separation regions.
- Transferred pressure-based loading into a structural FEA setup to study ramp deformation, stress concentration near hinge/bracket regions, reaction forces, and design risk areas.

Rocket Flight Numerical Simulation

2025

Programming & Algorithms Project | Python, NumPy, Matplotlib

- Simulated rocket altitude, velocity, acceleration, drag, apogee, and flight time using iterative time-step calculations, numerical integration logic, and desk checks for validation.

Rubber Band Powered Vehicle

2025

Engineering Graphics Final Project | SolidWorks, AutoCAD

- Designed vehicle components and produced ANSI-style engineering drawings with dimensions, constraints, and manufacturability considerations.

CERTIFICATIONS

Certified Associate in Project Management (CAPM), Project Management Institute | Earned May 2026

ENGINEERING PROJECT PORTFOLIO

Project evidence in one PDF for internship and research applications

Portfolio Focus

Mechanical engineering project work connecting aerospace inlet analysis, structural simulation, prosthetic hand design, Python numerical simulation, sensor data processing, and technical leadership/documentation.

CORE TECHNICAL AREAS

CAD	SolidWorks, AutoCAD, PTC Creo, Onshape, Fusion 360, assemblies, mates, drawings, tolerances
Simulation	ANSYS Fluent, ANSYS Mechanical, CFD, FEA, meshing, pressure contours, stress, deformation
Programming	Python, NumPy, Matplotlib, numerical simulation, data processing, plots, iterative methods
Leadership	Team co-lead, CAD training, technical documentation guide, workshop delivery, design communication

PROJECT INDEX

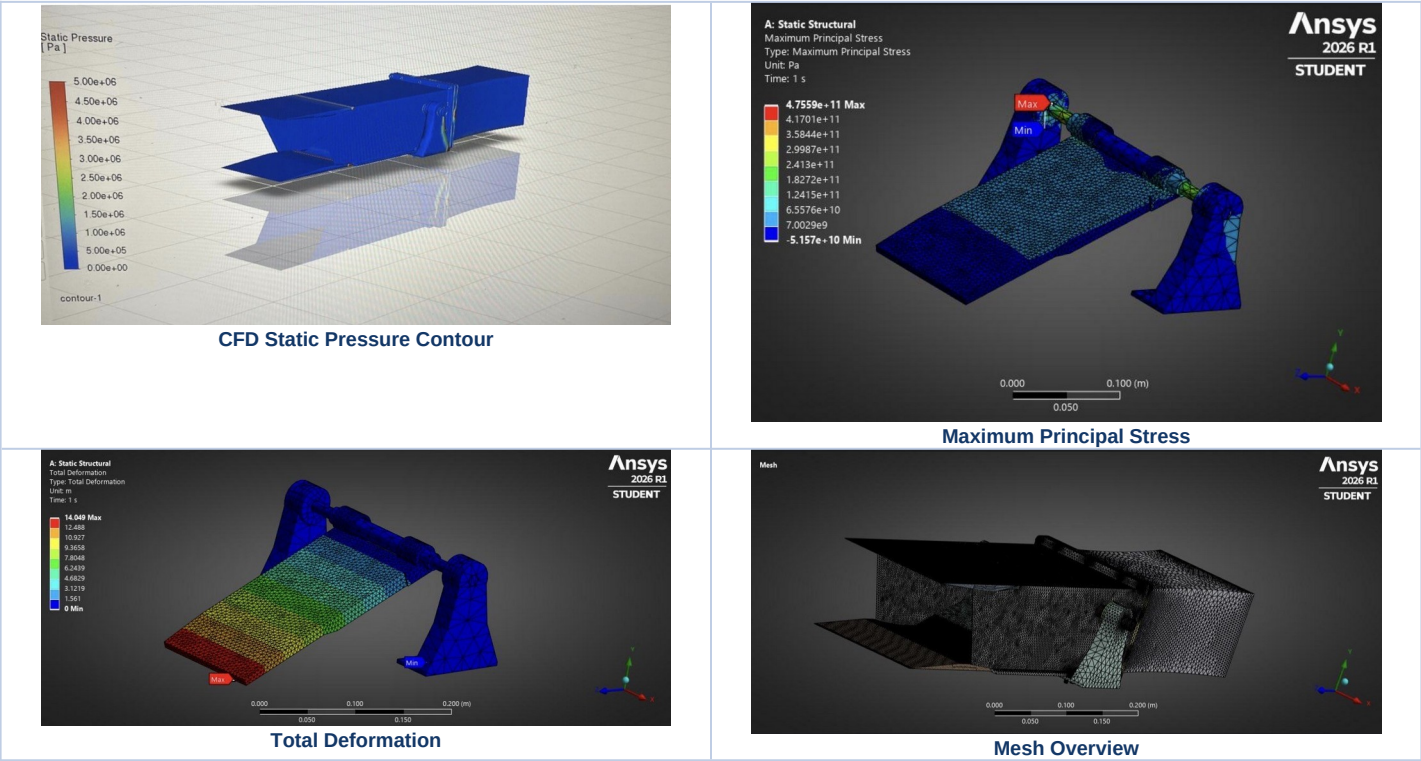
- Project 1 - Variable-Ramp Supersonic Inlet CFD & FEA Study
- Project 2 - Prosthetic Hand Design, Colegio Prosthetics Research (CPR)
- Project 3 - Rocket Flight Numerical Simulation
- Project 4 - Elevator Sensor Data Processing
- Project 5 - Rubber Band Powered Vehicle CAD Design

HOW TO READ THIS FILE

- Page 1 introduces the engineering project portfolio and project index.
- Pages 2-3 provide visual evidence and detailed project case studies.
- Page 4 summarizes additional engineering projects that support CAD, programming, and analysis skills.

PROJECT 1 | VARIABLE-RAMP SUPERSONIC INLET CFD & FEA STUDY

CFD and structural analysis for high-speed inlet design | Role: CAD modeling, CFD setup, FEA setup | Tools: SolidWorks, ANSYS Fluent, ANSYS Mechanical

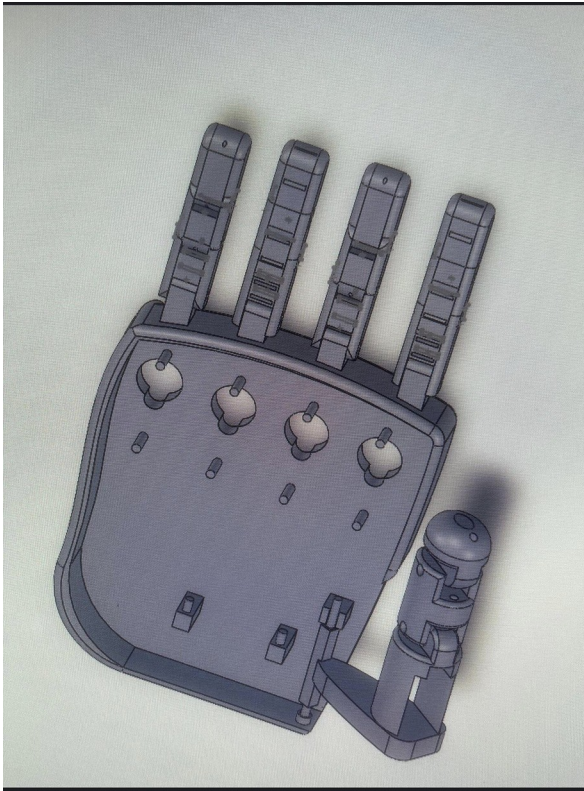


OBJECTIVE / PROCESS / OBSERVATIONS

Objective	Design and analyze a variable-ramp supersonic inlet concept and evaluate both aerodynamic loading and structural response.
Process	Modeled the inlet and ramp assembly, generated a computational mesh, evaluated pressure behavior with ANSYS Fluent, and used pressure-based assumptions in a static structural study.
Key Observations	Pressure behavior was assessed across the inlet geometry; stress concentrations appeared near hinge and support regions; maximum deformation occurred toward the free end of the ramp; results highlighted the importance of support design and mesh quality.
Skills Demonstrated	CAD modeling, meshing, CFD post-processing, structural analysis, design iteration, and engineering communication.

PROJECT 2 | PROSTHETIC HAND DESIGN - COLEGIO PROSTHETICS RESEARCH (CPR)

CAD design, assembly development, engineering drawings, and technical documentation | Role: Design Engineer / Co-Leader



Prosthetic Hand CAD Assembly

Objective

Support development of prosthetic hand components with emphasis on functionality, manufacturability, tolerance control, and realistic assembly integration.

Tools

SolidWorks, AutoCAD, Onshape, PTC Creo, Fusion 360

Skills Demonstrated

CAD modeling, assemblies, mates, tolerance design, engineering drawings, design-for-manufacturing, technical documentation, team leadership.

TECHNICAL CONTRIBUTIONS

- Designed and refined prosthetic finger and palm components in CAD to improve ergonomics, component fit, prototype realism, and manufacturability.
- Improved tolerances between components to support smoother assembly, reliable prototype integration, and future fabrication improvements.
- Built SolidWorks assemblies by defining mates, component constraints, and key measurements to verify interaction between moving parts.
- Created engineering drawings for selected components with dimensions, tolerances, and manufacturing details.
- Evaluated finger mechanism concepts to improve dexterity, range of motion, and functional movement.

LEADERSHIP

- Co-led a 7-member design/manufacturing team and supervised member work in CAD design, prototype development, and documentation.
- Led CAD skill-development workshops focused on SolidWorks modeling, assembly strategy, and design-for-manufacturing.
- Created an end-of-semester documentation guide to standardize research, design decisions, tolerances, errors found, and future improvements.

ADDITIONAL ENGINEERING PROJECTS

Supporting evidence of programming, sensor data analysis, CAD design, and engineering communication

Rocket Flight Numerical Simulation

Programming & Algorithms | Python, NumPy, Matplotlib

- Modeled rocket altitude, velocity, acceleration, drag, apogee, and total flight time using iterative time-step calculations.
- Compared idealized and drag-included behavior and used desk checks to validate the numerical workflow.
- Demonstrated programming fundamentals, numerical modeling, data visualization, and engineering interpretation.

Elevator Sensor Data Processing

Physics / Programming Project | Python, NumPy, Matplotlib, Phyphox sensor data

- Processed acceleration and pressure/velocity data from an elevator experiment using integration, differentiation, filtering, and bias correction.
- Created dashboards comparing acceleration, velocity, displacement, and pressure-derived motion trends.
- Demonstrated experimental data analysis, signal smoothing, drift correction, and technical explanation.

Rubber Band Powered Vehicle

Engineering Graphics Final Project | SolidWorks

- Designed a mechanical vehicle concept using SolidWorks part modeling and assembly practices.
- Prepared engineering drawings with dimensions and design constraints to communicate manufacturable geometry.
- Demonstrated CAD communication, mechanical design thinking, and early-stage prototype planning.

PROJECT EVIDENCE SUMMARY

- Main technical evidence: CFD pressure contour, FEA stress/deformation results, mesh overview, and prosthetic hand CAD assembly screenshots.
- Project range: aerospace analysis, biomechanical CAD design, Python numerical simulation, sensor data processing, and early-stage mechanical design.
- Document is structured as a project-only portfolio focused on technical evidence, CAD/CFD/FEA work, and engineering communication.